

ABHISHEK AGRAHARI

Embedded Software Engineer · EV IoT, Telematics & Vehicle Intelligence

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PROFESSIONAL SUMMARY

Embedded Software Engineer on the IoT team at Vecmocon, building the connected cards inside electric two- and three-wheelers — a BMS-integrated IoT card and fleet-management GPS devices on TI MCUs with Quectel EC200 cellular, streaming into VEC-TR and Battery Buddy (in-house fleet data platforms). Analyse telemetry from 20,000+ deployed devices; built a BLE app for EV scooter control and fleet management. Previously owned products end-to-end from schematic to commercial deployment: 2,000+ units validated, field issues cut by 90%. Strong in FreeRTOS, bootloaders/FOTA, low-power design and hardware bring-up.

TECHNICAL SKILLS

EV & Automotive	CAN / CAN-FD, BMS & VCU integration, motor-controller comms, fleet telematics & GPS tracking, AIS-140, FOTA / OTA
Microcontrollers	TI MCU, STM32 (F1 / F4), ESP32 / ESP32-S3, ARM Cortex-M, ATmega328P, 8051
Languages / RTOS	C, C++, Python, MATLAB / Simulink, Assembly · FreeRTOS, CMSIS-RTOS, embOS, bare-metal
Connectivity	UART, SPI, I2C, CAN, BLE, MQTT, HTTP, NB-IoT · Quectel EC200-series, Neoway N58 cellular modules
Cloud & Data	VEC-TR, Battery Buddy, ThingsBoard, MQTT telemetry pipelines, fleet data analysis
Firmware & Tools	Bootloaders, interrupt-driven design, DMA, low-power, memory optimisation · JTAG / SWD, logic analyser, oscilloscope, CAN analyser, STM32CubeIDE, Keil, Git, Jira

WORK EXPERIENCE

Embedded Software Engineer — IoT Team · Vecmocon Technologies May 2026 – Present · Noida

- Develop TI MCU firmware for multiple EV IoT cards — a BMS-integrated card and fleet-management GPS devices — each interfacing the vehicle's BMS, motor controller and VCU over CAN for OEM and fleet deployments.
- Integrate Quectel EC200-series cellular modules for real-time uplink; stream device telemetry into Battery Buddy and VEC-TR, Vecmocon's in-house server for fleet data storage and analytics.
- Built a BLE mobile app for EV scooter control and a fleet-management user interface that communicate directly with the IoT card.
- Analyse fleet telemetry from 20,000+ devices deployed in the market to surface field failure patterns and feed fixes back into firmware; developing application-layer code in MATLAB/Simulink and building AIS-140 compliance expertise.

Embedded Software Engineer · MLworkX Pvt. Ltd. Jun 2025 – May 2026 · Noida

- Led end-to-end R&D of production embedded products for global clients — schematic design through firmware architecture to commercial deployment — across STM32, ESP32 and 8051 platforms; integrated Neoway N58 cellular for real-time IoT communication.
- Led real-time debugging and validation (oscilloscope, logic analyser, JTAG/SWD) — reduced field issues by 90% and validated 2,000+ production units; optimised firmware for low power and owned board bring-up.

Embedded Software Engineer Intern · Surfytch Electronics Dec 2024 – May 2025 · Ghaziabad

- Developed C/C++ firmware on ATmega328P and STM32 with FreeRTOS; built a UART-based FOTA bootloader and a complete RFID access-control system; optimised interrupt handling and memory footprint.

KEY PROJECTS

- EV IoT Cards — BMS & Fleet GPS** (Vecmocon · TI MCU · Quectel EC200 · 2026) — Firmware for a family of automotive IoT cards: BMS/VCU/motor-controller acquisition over CAN, cellular uplink, GPS tracking with AIS-140-aligned behaviour, telemetry pipeline into VEC-TR and Battery Buddy. Production hardware in OEM vehicles.
- BLE App — EV Scooter Control & Fleet UI** (Vecmocon · 2026) — Mobile app talking directly to the IoT card over BLE with a defined command/response protocol for scooter control and diagnostics, plus a fleet-management interface for operators.
- Fleet Telemetry Analysis — 20K Devices** (Vecmocon · 2026) — Analysed field data from 20,000+ deployed devices in VEC-TR / Battery Buddy to identify recurring failure and connectivity patterns; findings fed into firmware priorities and validation cases.
- Vehicle Telematics Device** (STM32 · Neoway N58 GSM · CAN · Jul-Sep 2025) — Live vehicle data over CAN, telemetry to ThingsBoard via MQTT, FOTA over HTTP for remote firmware upgrades.
- Automatic Pill Dispenser** (Commercial product, US client · Nov 2025–Mar 2026) — STM32F103 firmware controlling 4 motors and multiple sensors; Android S720 board for UI. Delivered end-to-end, in commercial production.
- Smart Hip Trainer** (Wearable · ESP32-S3 · Jan–Mar 2026) — Owned schematic design and firmware; Bosch BNO055 IMU motion tracking streamed live to a companion app over BLE.

EDUCATION

B.Tech — Electronics & Communication Engineering
United College of Engineering and Research, AKTU Lucknow
Sept 2021 – May 2025 · CGPA 8.6 / 10.0 — Department Topper

Certifications: Design & Fabrication of Electronic Circuits ·
Innovation in Circuit Engineering using IoT · Springer Conference
Presentation · Letter of Appreciation

ACHIEVEMENTS

- Presented a live demo of the RedRaven™ fire detection system aboard **INS Airavat** to the Admiral and Head of Naval Dockyard, Visakhapatnam.
- Granted Indian Patent** — MIMO antenna design.
- Best Paper Award, ETMCIS 2024 (IEEE)** — Hybrid Metal-Graphene THz MIMO Antenna; 2 papers published in international journals/conferences.